

Forecasting and Sustainability in Middle Eastern Tourism, with Emphasis on Saudi Arabia: A PRISMA 2020 Systematic Review of the Empirical Literature, 2015–2026

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Abstract

Background. Tourism is a diversification pillar for Gulf economies, and Saudi Arabia in particular, yet it is unclear whether the empirical literature that forecasts tourism and the literature that studies its sustainability inform one another. **Objectives.** To map and synthesise empirical studies of (A) tourism or hospitality forecasting, (B) tourism-demand modelling, and (C) quantitative tourism–sustainability analysis in Saudi Arabia and the wider Middle East, and to identify evidence gaps. **Data sources.** Crossref REST API, last searched 5 October 2026 (journal articles, 2015–2026; 70 queries; 7,000 records, 2,237 unique), plus targeted title searches seeded by web-search leads. **Eligibility and appraisal.** Peer-reviewed quantitative studies in English set in the Middle East (MENA, GCC, Türkiye, Iran, Israel). Screening was by a single reviewer (an AI agent) at title/abstract level; full texts were retrieved for 22 of 43 provisionally included studies and checked for eligibility (1 excluded). Risk of bias was appraised on full texts with seven-item checklists for forecasting (F1–F7) and econometric (E1–E7) studies. Results were synthesised descriptively, by keyword co-occurrence analysis and by thematic synthesis; no meta-analysis was performed. **Results.** 42 studies were included (16 forecasting, 11 demand-modelling, 15 sustainability-nexus); 38 appeared from 2020 onward. Saudi Arabia was the setting of 13 studies but only 4 forecasting studies. Of 16 forecasting studies, only 1 addressed sustainability; all 26 non-forecasting studies used econometric or statistical designs, only 1 projected outcomes forward (climate-scenario simulation), and none evaluated out-of-sample forecast accuracy. Within the retrieved corpus the two literatures are therefore largely disconnected, and the split persisted in three sensitivity analyses. Among the 11 forecasting studies assessed at full text, none tested accuracy differences statistically, 1 compared against a naïve benchmark and 1 reported prediction intervals. Emission effects of tourism were mixed in sign (two increasing, two decreasing, one mixed, one ambiguous), and no study stated whether aviation emissions were included. **Conclusions.** Forecasts are rarely linked to environmental or capacity outcomes, and sustainability analyses are mostly ex post. We outline a capacity-aware forecasting agenda. Findings are bounded by retrieval caps, abstract-level coding and single-reviewer screening, and must be confirmed against full texts before submission. **Funding.** None declared [to be completed by the authors]. **Registration.** Not registered.

Keywords: tourism demand forecasting; sustainability; Saudi Arabia; Vision 2030; Middle East; machine learning; PRISMA 2020

1 Introduction

Tourism is central to the economic diversification plans of Gulf states. Saudi studies frame tourism research around Vision 2030 and diversification away from oil dependency (Alamoudi

and Abed, 2020; Abuhulaibah, 2026), and recent work links tourism growth to climate and environmental outcomes in the Kingdom (Chouari, 2025; Alfehaid, 2026; Raihan et al., 2025). Two methodological traditions are relevant. The first is tourism demand forecasting, which has been reviewed repeatedly in global terms (Song et al., 2019; Peng et al., 2014; Jiao and Chen, 2018) and has moved from time-series and econometric models toward machine learning, deep learning and internet-search data (Law et al., 2019; Zhang et al., 2020b). General forecasting competitions have emphasised out-of-sample evaluation against strong benchmarks (Makridakis et al., 2020, 2022). The second tradition is the quantitative analysis of tourism’s economic, social and environmental consequences, typically through cointegration and panel designs.

Saudi-focused work has examined tourism’s contribution to Vision 2030 and SDG 8 (Iqbal and Aftab, 2025), but within the sources retrieved here we did not identify a PRISMA-guided review that examines *forecasting* and *sustainability* evidence together for Saudi Arabia and the Middle East. This matters because forecasts are used to plan capacity and investment, whereas sustainability outcomes depend on that capacity. If the two literatures do not meet, forecasts may be accurate yet blind to environmental limits, and sustainability analyses may be rigorous yet unable to anticipate future pressure.

Objectives. Following PRISMA 2020 (Page et al., 2021a,b), we ask: **RQ1** What forecasting, demand-modelling and tourism–sustainability studies exist for Saudi Arabia and the Middle East, and what are their settings and methods? **RQ2** How rigorously are forecasting studies evaluated, as visible in their reporting? **RQ3** To what extent do forecasting studies incorporate sustainability, and do sustainability studies produce forecasts? **RQ4** What research agenda follows?

Contributions relative to prior reviews. Section 2 compares this review with the closest of 25 prior reviews whose bibliographic records were verified in Crossref. Against that set, the review contributes: (i) a joint, PRISMA-structured map of *forecasting* and *quantitative sustainability* evidence for Saudi Arabia and the Middle East, a scope that, among the reviews we located, is covered only in parts (global forecasting reviews; global tourism–carbon and carrying-capacity reviews; regional reviews of GCC or Saudi tourism without a forecasting lens); (ii) a cross-classification showing that the two literatures are largely disconnected in this region, together with a typology of the *direction* of the tourism–environment links that are studied; (iii) a regional comparison of these findings with the global reviews; and (iv) a research agenda positioned against the existing carrying-capacity early-warning literature. Contribution (i) is a scope contribution, and (ii)–(iii) are synthesis contributions; we do not claim methodological novelty in review procedure.

2 Literature Background and Related Reviews

We searched Crossref for prior reviews with 24 logged queries (review, bibliometric, meta-analysis and scoping-review terms combined with forecasting, sustainability, carbon, carrying capacity, Saudi Arabia, GCC and Middle East; `prior/prior_search_log.json`) and checked additional regional reviews by web search. This identified 25 closely related reviews; all are listed with their open-access status in the repository (`prior/, data/oa_status.json`). To keep the reference list mostly open access, Table 1 shows the open-access reviews together with the three closed-access reviews that the novelty argument depends on (Jiao and Chen, 2018; Sun et al., 2022; Saleh et al., 2021). Omitting the other closed-access reviews does not change the conclusions below: none of them, judged from abstract or title, covers forecasting and sustainability together for this region. Scope is coded from the Crossref abstract where one is deposited and otherwise from the title; two entries rely on web-search snippets and are marked.

Global forecasting reviews synthesise models, data sources and accuracy (Peng et al., 2014; Jiao and Chen, 2018; Song et al., 2019; Zhang et al., 2020a; Dowlut and Gobin-Rahimbux, 2023; Wu et al., 2024; Henriques and Nobre Pereira, 2024). Where abstracts are available, none indicates a regional focus on the Middle East or a sustainability dimension.

Tourism–sustainability reviews address carbon emissions (Sun et al., 2022), automation for sustainability (Majid et al., 2023) and carrying capacity (Long et al., 2022; Ajuhari et al., 2023). The closest is Sun et al. (2022), a systematic review of econometric tourism–emission studies, which overlaps our Theme S in method and outcome but is global. The carrying-capacity reviews are the only ones that connect sustainability to forecasting: Long et al. (2022) call for improved dynamic forecasting and early warning of tourism environmental carrying capacity, and primary studies already forecast early-warning indices (e.g. Ye et al., 2020).

Regional reviews cover GCC tourism research (Saleh et al., 2021), Saudi tourism and SDG 8 (Iqbal and Aftab, 2025) and ESG in Saudi hotels (Alhejaili and Ahmad, 2025). The closest is Saleh et al. (2021), a PRISMA review of 23 GCC tourism studies, which found most to focus on Dubai and on tourism planning; its abstract does not indicate coverage of forecasting.

Closest prior work and difference. The two reviews most dangerous to our novelty claim are Saleh et al. (2021) (same region and reporting standard) and Sun et al. (2022) (same sustainability outcome and study designs). This review differs from both in jointly coding forecasting and sustainability studies and analysing their connection, in extending the region to the Middle East with a Saudi focus, and in covering 2015–2026, including the post-2021 growth in Saudi studies. It is narrower than both in one respect, as it rests on abstract-level coding (Section 6.1). We therefore state its novelty conservatively: among the reviews we located, none jointly synthesises forecasting and quantitative sustainability evidence for this region.

Table 1: Closest prior reviews compared with this review (open-access reviews plus the closed-access reviews essential to the comparison). F = forecasting; S = sustainability; F↔S = analyses links between forecasting and sustainability. ✓ = indicated; – = not indicated; “title” = abstract not deposited in Crossref, scope coded from the title only; “snippet” = scope from web-search snippet (unverified). Access from Unpaywall (gold, hybrid, green, bronze = open access).

Review	Type, sample	Coverage	Region	F	S	F↔S	Scope coded from	Access
<i>Global forecasting reviews</i>								
Peng et al. (2014)	Meta-analysis	n/r	Global	✓	–	–	title	green
Jiao and Chen (2018)	Review, 72 studies	2008–2017	Global	✓	–	–	abstract	closed
Song et al. (2019)	Curated review	n/r	Global	✓	–	–	title	bronze
Zhang et al. (2020a)	Knowledge mapping	n/r	Global	✓	–	–	title	green
Dowlut and Gobin-Rahimbux (2023)	SLR (deep learning)	resort hotels	Global	✓	–	–	title	gold
Wu et al. (2024)	Systematic (big data)	n/r	Global	✓	–	–	abstract	hybrid
Henriques and Nobre Pereira (2024)	PRISMA SLR, 20	Scopus, WoS	Global	✓	–	–	abstract	gold
<i>Tourism–sustainability reviews</i>								
Sun et al. (2022)	Systematic, 81 econometric studies	2013–2021	Global	–	✓	partial (decarbonisation scenarios)	snippet	closed
Majid et al. (2023)	Systematic	automation	Global	–	✓	–	title	hybrid
Long et al. (2022)	Bibliometric, 297	carrying capacity	Global	partial	✓	calls for forecasting/early warning	abstract	gold
Ajuhari et al. (2023)	SLR, 100	carrying-capacity methods	Global	–	✓	–	abstract	gold
<i>Regional reviews</i>								
Saleh et al. (2021)	PRISMA SR, 23	GCC tourism	GCC	–	partial (impacts)	–	abstract	closed
Iqbal and Aftab (2025)	SLR	Vision 2030, SDG 8	Saudi Arabia	–	✓	–	snippet	hybrid
Alhejaili and Ahmad (2025)	Literature review	ESG, 2020–2024	Saudi hotels	–	✓	–	abstract	gold
This review	PRISMA SR, 42 studies	2015–2026, Crossref	Middle East, Saudi focus	✓	✓	✓ (cross-classified)	abstract	–

3 Methodology

The review follows the PRISMA 2020 statement; the completed checklist, with honest status per item, is in Appendix A. Deviations are reported explicitly below and in Section 6.1.

3.1 Review Protocol and Registration

No protocol was registered. The inclusion criteria, search design and coding scheme were fixed before the eligibility stage and are archived with the code and data (Appendix D). Registration on a general registry such as OSF is recommended before journal submission.

3.2 Information Sources

The primary source was the Crossref REST API (accessed 5 October 2026), filtered to journal articles published 1 January 2015 to 31 December 2026. Crossref was chosen because it was the only bibliographic index reachable from the execution environment; Scopus, Web of Science and Google Scholar were not accessible. Four earlier web-search screening passes (*search snippets only*) produced candidate leads. These leads were not counted as identified records; they seeded targeted Crossref title searches (“other methods”), and every record that was retained had its metadata re-verified by DOI lookup in Crossref. Publisher full texts could not be retrieved directly because the environment’s network policy returned errors for publisher and DOI-resolver hosts; open-access copies were obtained from repositories (Section 3.5). At revision, a supplementary search applied the same concept $\times$ location design to Semantic Scholar (50 queries; journal articles, 2015–2026; `search/extra_db_search.py`); its new records were screened with the same rules (Section 3.5). OpenAlex was also queried but its shared free quota was exhausted from the review environment, and Scopus and Web of Science require institutional access; both are listed as planned amendments in the protocol (`prisma/OSF_protocol.md`).

3.3 Search Strategy

We ran 70 queries using the `query.bibliographic` parameter with `rows=100`: eight concept strings (e.g. “tourism demand forecasting”, “tourism CO2 emissions”, “sustainable tourism modelling”) crossed with four regional terms (Saudi Arabia, Middle East, GCC, Hajj/Umrah), plus “tourism demand forecasting” and “tourism sustainability” for 19 additional location terms (Appendix C). Each query returned its 100 top-ranked records, so retrieval is capped and not exhaustive (Crossref’s reported totals count fuzzy matches across all works and are not counts of relevant records) (Section 6.1). The full list, dates and reported totals are in Appendix C and `data/search_log.json`. PRISMA-S was not formally applied; the limits of relying on a single index are discussed by Gusenbauer and Haddaway (2020).

Table 2 summarises the strategy.

3.4 Inclusion and Exclusion Criteria

Studies were eligible if they: (i) were peer-reviewed journal articles, 2015–2026; (ii) were reported in English; (iii) were set in the Middle East, defined here as Saudi Arabia, the other GCC states, MENA countries, Türkiye, Iran, Iraq and Israel (multi-country panels were eligible when Middle Eastern units were analysed); and (iv) were quantitative empirical studies falling under one of three themes: **F** – forecasting of tourism or hospitality outcomes (arrivals, demand, spending, revenue, hotel price or occupancy); **D** – econometric or statistical modelling of tourism demand or outcomes without a forecasting objective; **S** – quantitative analysis of tourism together with a sustainability

Table 2: Search strategy: databases, search strings and filters (full query log in Appendix C).

Element	Specification
Database	Crossref REST API (<code>query.bibliographic</code>); accessed 5 October 2026
Concept block (8 strings)	“tourism demand forecasting”, “tourist arrivals forecasting”, “tourism forecasting machine learning”, “tourism forecasting deep learning”, “hotel occupancy forecasting”, “tourism sustainability prediction”, “tourism CO2 emissions”, “sustainable tourism modelling”
Location block 1 (crossed with concepts)	Saudi Arabia; Middle East; GCC; Hajj Umrah
Location block 2 (with “tourism demand forecasting” and “tourism sustainability”)	Gulf Cooperation Council; GCC; Middle East; MENA; Dubai; UAE; United Arab Emirates; Qatar; Oman; Bahrain; Kuwait; Jordan; Egypt; Turkey; Iran; Morocco; Tunisia; Lebanon; Israel
Filters	Document type = journal article; publication date 1 January 2015 – 31 December 2026; top 100 records per query
Other methods	Targeted Crossref title searches seeded by web-search leads; DOI verification of every retained record

dimension (environmental, social or economic-sustainability variables). Excluded were studies that examined only the tourism–economic-growth nexus without a demand or sustainability outcome, perception or structural-equation surveys, qualitative or conceptual work, operational prediction such as crowd control or clinical outcomes during pilgrimage, non-journal records, and records whose region or tourism link could not be determined from the available metadata.

The criteria are summarised in Table 3.

3.5 Study Selection

Selection had four stages. *Stage 1* (automated): deduplication by DOI, then a rule-based filter requiring a tourism term, a Middle East location term and a forecasting or sustainability term in the title or abstract (`search/screen.py`). *Stage 2* (automated): a method-term filter (forecast, predict, time series, ARDL, regression, machine learning and similar; `search/screen_stage2.py`). *Stage 3* (manual): title/abstract/metadata eligibility assessment of the remaining records, using the criteria above, with a reason recorded for every exclusion (Appendix E). One reviewer, an AI agent, performed Stage 3; there was no independent duplicate screening. *Stage 4* (full text): open-access full texts of the provisionally included studies were retrieved where the network policy allowed (directly, from Internet Archive copies, Semantic Scholar or Europe PMC; `search/oa_download*.py`) and checked against the same criteria. 22 of 43 were retrieved; 1 was excluded (a Hajj–GDP causality study without a sustainability variable; Triki, 2019) and 21 were confirmed. The 21 studies whose full texts could not be retrieved (11 closed access, 10 on blocked hosts) remain included on title-and-abstract evidence and are flagged as pending. A second-pass check of all exclusions coded as tourism–growth nexus only found one error: ? includes CO₂ emissions as an explanatory variable and therefore meets the Theme S criterion; it was re-included and confirmed at full text. To address reviewer independence, all 98 title/abstract records were exported in random order, without the first reviewer’s decisions, for blind screening by an independent human reviewer, with Cohen’s κ computed by `search/kappa.py` (`prisma/second_reviewer/`); this step is pending at the time of writing.

Table 3: Inclusion and exclusion criteria.

Criterion	Inclusion	Exclusion
Publication type	Peer-reviewed journal article	Conference proceedings, book chapters, theses, preprints
Period	2015–2026	Before 2015
Language	English	Other languages
Setting	Saudi Arabia, other GCC states, MENA, Türkiye, Iran, Iraq, Israel; multi-country panels with Middle Eastern units	No Middle Eastern unit analysed
Design	Quantitative empirical study	Perception or SEM surveys, qualitative, conceptual or review papers
Topic	F: forecasting a tourism or hospitality outcome; D: modelling tourism demand or outcomes; S: tourism with an environmental, social or economic-sustainability variable	Tourism–growth nexus only; crowd, health or operational prediction; non-tourism outcomes
Information	Region and tourism link determinable from metadata or abstract	Region or tourism link not determinable

3.6 Data Extraction

For every study with a retrieved full text (21 of 42), data were extracted from the full text with a written protocol (`search/ft_extraction_protocol.md`) that requires a verbatim evidence quote for every item and forbids inference (`data/ft_extract_*.json`). Extraction was performed by AI agents working in parallel and spot-checked against the PDFs by the lead reviewer. Forecasting items were the holdout design, horizon, forecast origin, comparison models, best model, reported accuracy and ex-ante forecasts; econometric items were the estimator, tourism variables, coefficients with sign and significance, whether aviation emissions were included, and any out-of-sample forecast. For the remaining studies, the first reviewer coded items from the abstract and bibliographic record (`data/included.json`): theme; country; outcome; data period and frequency; method family (classical statistical or econometric, machine or deep learning, hybrid, ensemble or multi-family comparison, grey); presence of a benchmark comparison; reporting of an accuracy metric; presence of a forward projection; and sustainability dimension. Items not visible in the abstract were coded “n/r” and never inferred. Open-access status was recorded for every included study with Unpaywall (`data/oa_status.json`): 31 of the 42 are open access and 11 are closed access. Access did not influence eligibility, because excluding closed-access studies would bias a systematic review. For 8 of the 42 included records, Crossref held no abstract; these were coded from title and metadata only and are flagged in the tables.

The extraction form is shown in Table 4.

3.7 Quality Appraisal

Full-text studies were appraised with two seven-item checklists adapted from PROBAST (Wolff et al., 2019) and forecast-evaluation practice (Hyndman and Athanasopoulos, 2021; Diebold and Mariano, 1995). For forecasting studies: F1 holdout period reported; F2 naïve or seasonal-naïve benchmark; F3 scale-free or relative accuracy measure; F4 statistical test of accuracy differences; F5 no information leakage (predictors known at the forecast origin); F6 horizon specified; F7 prediction intervals. For econometric studies: E1 unit-root tests; E2 cointegration or long-run test; E3 structural breaks; E4 cross-sectional dependence or slope heterogeneity (panels only); E5 endogeneity or omitted variables addressed; E6 residual diagnostics or stability; E7 data source and period. Each item is rated Yes, No or Unclear with an evidence quote. Studies without a full

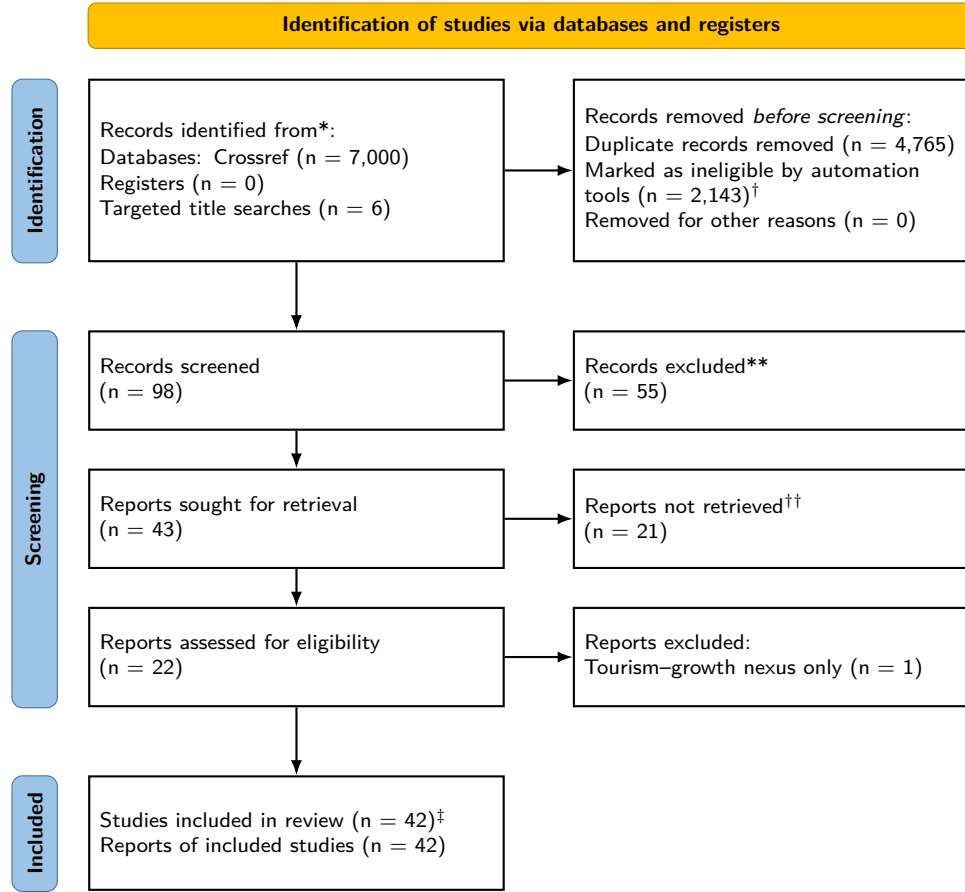


Figure 1: PRISMA 2020 flow diagram, drawn after the official template for new systematic reviews (Page et al., 2021a). *Crossref was the only database searched (70 queries, top 100 records each; Appendix C); no registers were searched. Duplicates include the 2 targeted-search records already retrieved from Crossref. **Excluded by one reviewer (an AI agent) on title and abstract: not a quantitative forecasting, demand or sustainability study (33); tourism-growth nexus only (8); outside the Middle East (5); operational or health prediction (4); region or tourism link unconfirmed (3); non-English or proceedings (2); listed in Appendix E. [†]Keyword filters flagged 2,145 records; 2 were retained because targeted searches also identified them. ^{††}11 closed access and 10 open access on hosts blocked from the review environment. [‡]21 studies confirmed at full text and 21 included on title-and-abstract evidence pending full-text retrieval.

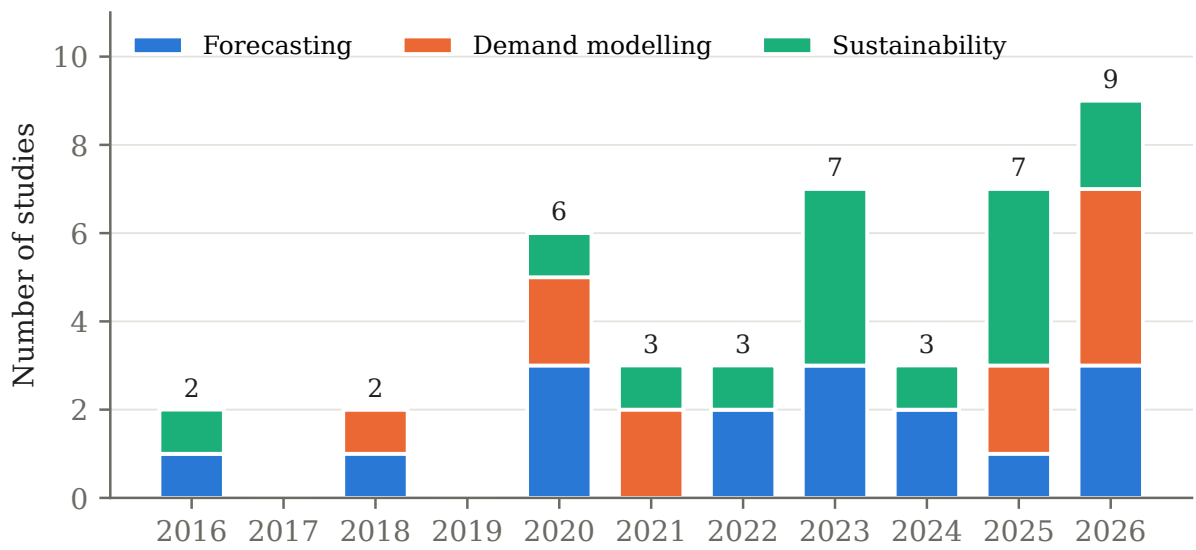
text were assessed only with the abstract-level reporting screen described in Section 4.7. No study was excluded on quality grounds.

3.8 Data Analysis

Following the competing reviews, the analysis has three parts. (i) A descriptive analysis of publication years, journals, settings, method families and accuracy measures (Figures 2–7). (ii) A keyword co-occurrence analysis: because author keywords are not deposited in Crossref, we applied a controlled vocabulary of 28 terms (`search/make_figures.py`) to titles and abstracts, retained terms occurring in at least three studies, linked terms co-occurring in at least two, and detected clusters by greedy modularity maximisation (Figure 5). (iii) A thematic synthesis by theme (F, D, S), followed by a cross-theme analysis; for Theme S the direction of the reported tourism effect on the environmental outcome is vote-counted. (iv) Sensitivity analyses of the cross-theme split: S1 adding back the growth-nexus exclusions, S2 excluding Türkiye, and S3 restricting to full-text-confirmed studies (`search/sensitivity.py`). No meta-analysis or effect-size pooling was attempted because outcomes, horizons, data scales and metrics are heterogeneous. Reporting-bias and certainty-of-evidence assessments (PRISMA items 14, 15, 21, 22) were not performed.

Table 4: Data extraction form (abstract-level; coding scheme in Appendix B).

Item	Values
Bibliographic	Authors, year, journal, volume, pages, DOI (Crossref); open-access status (Unpaywall)
Theme	F (forecasting), D (demand modelling), S (sustainability)
Setting	Country or multi-country panel
Outcome	e.g. arrivals, spending, revenue, hotel price or occupancy, CO ₂ , income equality
Data	Period and frequency
Method family	Classical; machine or deep learning; hybrid, ensemble or multi-family comparison; grey
Evaluation (F only)	Benchmark comparison; accuracy measure named; forward projection or scenario
Sustainability	Dimension stated (environmental, social, economic-sustainability, climate) or framing only

**Figure 2:** Year-wise distribution of the 42 included studies by theme. Labels give annual totals.

4 Results

4.1 Overview of the Included Studies

The 70 Crossref queries returned 7,000 records, of which 4,763 were duplicates, leaving 2,237 unique records. Four further records came from targeted searches and two others, found by those searches, had already been retrieved but removed by automated screening; 2,241 records were therefore screened and 2,145 were removed by the automated stages. Of 98 records screened on title and abstract, 55 were excluded, leaving 43 provisionally included studies. Full texts were retrieved for 22 of them; one was excluded at full text, leaving 42 included studies (Section 3.5, Figure 1). Exclusion reasons were: not a quantitative forecasting, demand or sustainability study (33); tourism–growth nexus only (8); outside the Middle East (5); operational, crowd or health prediction (4); region or tourism link unconfirmed (3); and non-English or proceedings (2). The two records reinstated after automated exclusion show that the automated filters have imperfect recall (Section 6.1).

The 42 included studies span 2016–2026; 38 appeared from 2020 and 16 from 2025 (Figure 2). They appeared in 30 journals; *Sustainability* published 10, *Tourism Economics* 3 and *Forecasting* 2, and the remaining 27 appeared in different journals (Figure 3). Saudi Arabia was the most

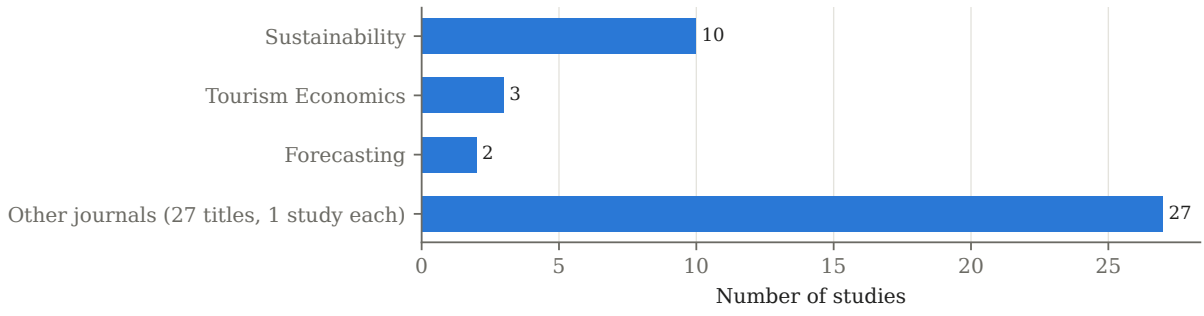


Figure 3: Journals publishing the included studies. Journals with a single included study are pooled.

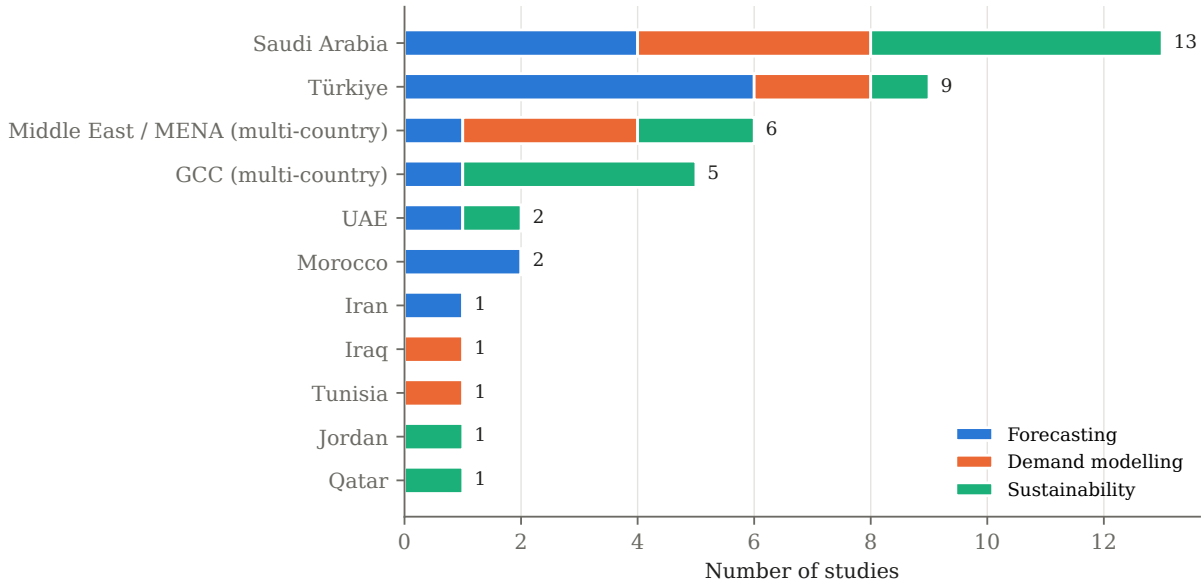


Figure 4: Study setting by theme. Multi-country panels are shown separately; the study covering Saudi Arabia, Türkiye and Italy is counted under Saudi Arabia.

frequent setting (13 studies), followed by Türkiye (9), multi-country Middle East or MENA panels (6) and GCC panels (5) (Figure 4). Forecasting studies concentrate on Türkiye (6) and Saudi Arabia (4); sustainability studies on Saudi Arabia and GCC panels.

4.2 Keyword Co-occurrence Analysis

The co-occurrence network has 24 terms and 104 links (Figure 5). Modularity clustering separates two groups. The first (blue) combines forecasting, tourism demand, tourist arrivals, ARIMA/SARIMA, machine and deep learning, search data, seasonality, COVID-19 and Türkiye. The second (orange) combines sustainability, CO₂ emissions, energy, EKC, economic growth, ARDL/cointegration, causality, panel data, religious tourism, Vision 2030, GCC and Saudi Arabia. Only 24% of co-occurrence weight crosses the two clusters, and the strongest bridges pass through *tourist arrivals*, which sustainability studies use as an explanatory variable rather than as a forecast target. The term *forecasting* co-occurs with *sustainability* in only two studies and with CO₂, energy or EKC terms in fewer than two. The network thus anticipates the thematic finding in Section 4.6.

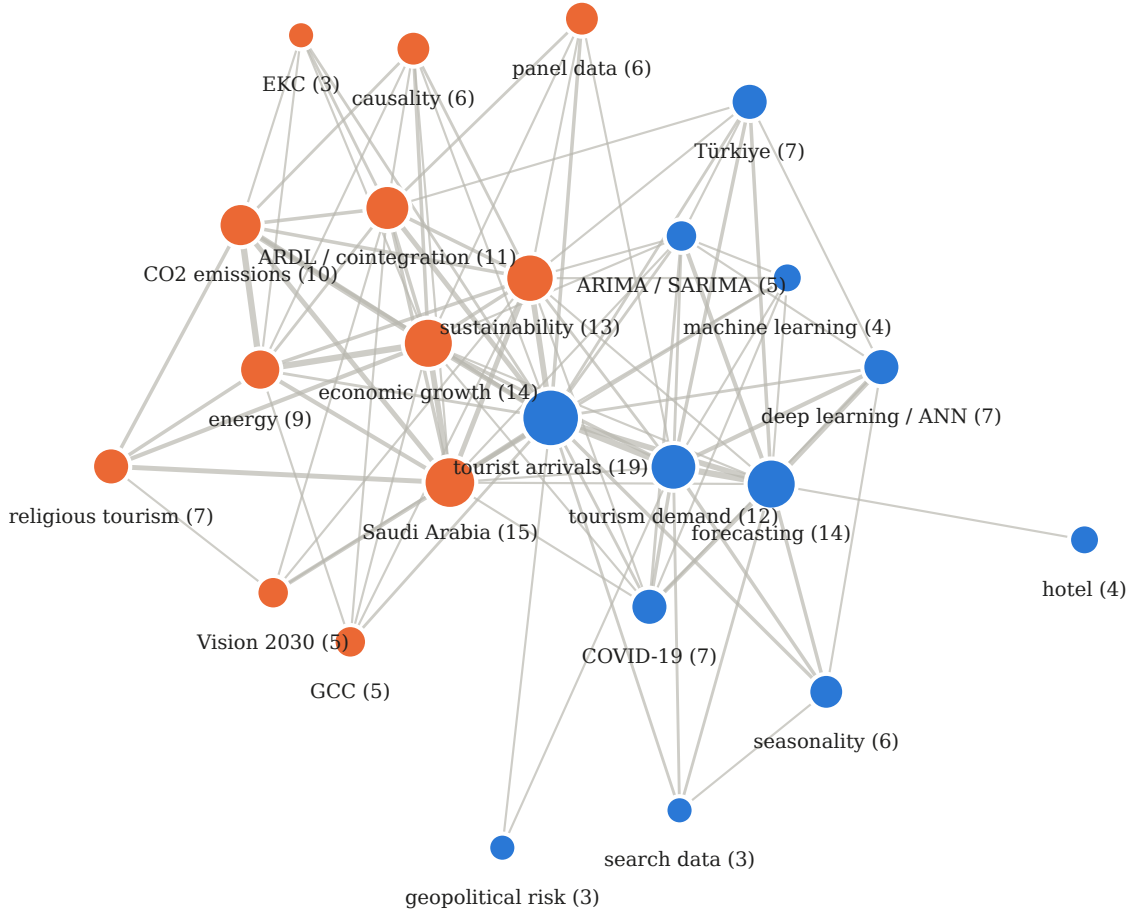


Figure 5: Keyword co-occurrence network of the included studies (controlled vocabulary applied to titles and abstracts). Node size reflects the number of studies containing the term (in brackets); line width reflects co-occurrence. Colours show the two modularity clusters: forecasting (blue) and sustainability–econometrics (orange).

4.3 Theme 1: Forecasting Studies

16 studies forecast a tourism or hospitality outcome (Table 5). By method family (Figure 6), 6 used machine or deep learning, 4 classical models, 5 hybrid or ensemble designs and 1 grey model. Of the 14 studies whose abstracts were available, 13 compared several models or benchmarks, which is encouraging. 13 abstracts named an accuracy metric, most often MAPE (5) and RMSE (3) (Figure 7), but only 6 produced a forward projection or scenario. Saudi forecasting evidence is thin: 4 studies, covering annual Umrah visitors to 2030 (Alamoudi and Abed, 2020), spending and ARIMA projections to 2030 (Louati et al., 2024), a machine-learning demand model with a geopolitical-instability proxy (Zamzami, 2026), and a machine-learning growth model whose abstract was unavailable (Alsulami et al., 2026). GCC hotel price forecasting with deep learning (Al Shehhi and Karathanasopoulos, 2020) and MENA hotel prices (Al Shehhi and Karathanasopoulos, 2018) are outside the arrivals-demand focus but within Theme F. Search-data forecasting of Dubai tourism (Rashad, 2022) and Marrakech arrivals (Laaroussi et al., 2023) indicate that internet-search predictors, studied globally (Wu et al., 2024), have reached the region.

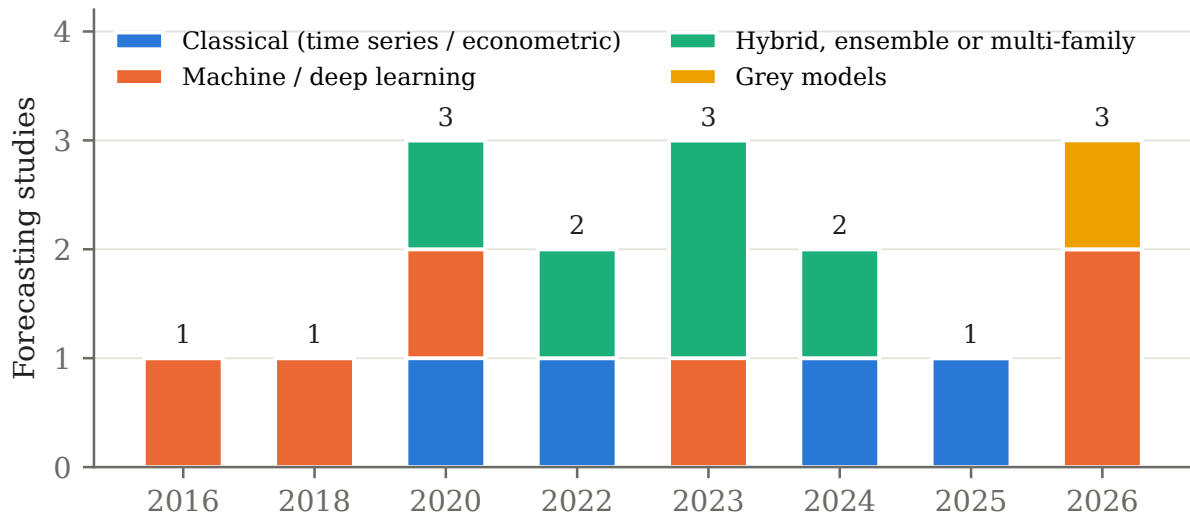


Figure 6: Forecasting studies by method family and publication year ($n = 16$).

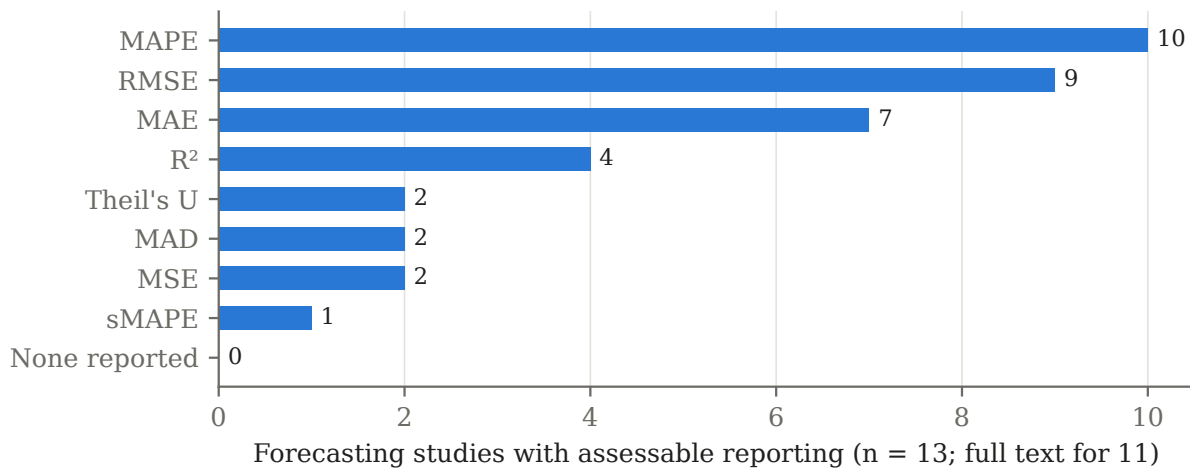


Figure 7: Accuracy measures named in the abstracts of forecasting studies. A study can name several measures.

Table 5: Forecasting studies (Theme F). “Benchmark”: named comparison of models or benchmarks in the abstract; “n/r”: not reported or abstract unavailable. Coded from abstracts and metadata only.

Study	Setting	Outcome	Data	Method family	Benchmark	Forward projection
Al Shehhi and Karathanasopoulos (2020)	GCC cities	hotel room prices	n/r (abstract unavailable)	ML/DL	n/r	n/r
Ahmadian (2026)	Iran	inbound tourist arrivals	2020-2023 (small sample)	Grey	Yes	Yes (2024-26)
Al Shehhi and Karathanasopoulos (2018)	MENA cities	hotel prices	n/r (abstract unavailable)	ML/DL	Yes	No
Ouassou and Taya (2022)	Morocco	regional tourist arrivals	annual, 1999-2018	Hybrid/ensemble	Yes	No
Laaroussi et al. (2023)	Morocco (Mar-rakech)	tourist arrivals	monthly	Hybrid/ensemble	Yes	No
Louati et al. (2024)	Saudi Arabia	tourist spending; economic landscape	2015-2021	Hybrid/ensemble	Yes	Yes (2022-30)
Alsulami et al. (2026)	Saudi Arabia	tourism growth	n/r (abstract unavailable)	ML/DL	Yes	Yes (2024-2034 projections; VotingR)
Zamzami (2026)	Saudi Arabia	tourism demand (tourist numbers)	2015-2024	ML/DL	Yes	No
Alamoudi and Abed (2020)	Saudi Arabia (Makkah)	international Umrah visitors	annual, 33 years	Classical	Yes	Yes (2020-2030: 7,628,163 (2020) to)
Cankurt and Subaşı (2016)	Türkiye	tourism demand	n/r (abstract unavailable)	ML/DL	n/r	n/r
Çuhadar (2020)	Türkiye	tourism revenues	n/r	Hybrid/ensemble	Yes	No
Tuncsiper (2023)	Türkiye	tourist numbers; tourism revenue	monthly, 2008-2022	ML/DL	No	No
Kayral et al. (2023)	Türkiye	tourist arrivals; tourism income	n/r	Hybrid/ensemble	Yes	Yes (2022/Q1-2022/Q4 under war / no)
Kurtulay and Kızılırmak (2024)	Türkiye	international visitor demand	monthly, 2002-01 to 2023-08	Classical	Yes	No
Bilek (2025)	Türkiye	tourism demand	monthly, 2008-2024	Classical	Yes	Yes (monthly forecasts 2025-2035 fr)
Rashad (2022)	UAE (Dubai)	tourism demand (travel search data)	monthly, 2019-01 to 2022-04	Classical	Yes	No

4.4 Themes 2 and 3: Demand-Modelling and Sustainability Studies

The remaining 26 studies are all classical econometric or statistical designs (ARDL, VECM, panel, gravity or spatial models; 26 of 26) and none evaluates out-of-sample forecast accuracy. One, Chouari (2025), simulates Saudi tourist flows under future climate scenarios (RCP 4.5 and 8.5), the closest example in the corpus to environment-aware projection; its link runs from climate to demand rather than from visitor numbers to environmental load. Theme S studies (15) mainly relate tourism to CO₂ emissions or environmental quality (13 of 15 include an environmental outcome), for example in GCC panels (Farooq et al., 2023b; Zmami and Ben-Salha, 2024), Middle East panels (Voumik et al., 2023; Bin Surayhid et al., 2026), the UAE (Majumdar and Paris, 2022), Türkiye (Bahar and Demir, 2023), and Saudi pilgrimage tourism (Naseem, 2025; Raihan et al., 2025). Social outcomes appear less often, for example income equality in the GCC (Alnafisah, 2025). Demand-modelling studies address determinants such as oil prices (Kisswani et al., 2020; Mabrouk and Alanzi, 2026), geopolitical risk (Hamida and Aloui, 2025), institutional quality (Wada, 2021) and, for Saudi Arabia, climate (Chouari, 2025; Abuhulaibah, 2026) and regional infrastructure (Alfehaid, 2026). Table 6 lists them.

Table 6: Demand-modelling (D) and sustainability (S) studies. All are econometric or statistical; none evaluates out-of-sample forecast accuracy (one simulates climate scenarios).

Study	Theme	Setting	Outcome	Data	Sustainability dimension
Shmoto (2026)	D	Iraq (Karbala)	hotel occupancy seasonality	monthly, 5 years, 27 hotels	none
Kisswani et al. (2020)	D	MENA (panel)	tourism receipts	n/r	none
Wada (2021)	D	MENA (panel)	international arrivals	2012-2018	none
Rafei Darani and Asghari (2018)	D	Middle East (panel)	international tourism demand	annual panel, 1995-2013	none
Chouari (2025)	D	Saudi Arabia	regional/seasonal tourist flows vs climate	2005-2024	environmental (climate)
Abuhulaibah (2026)	D	Saudi Arabia	tourism development	annual, 1995-2024	environmental (climate vulnerability)
Alfehaid (2026)	D	Saudi Arabia	regional tourism spending	13 regions, 2024	environmental (green infrastructure), SDG
Mabrouk and Alanzi (2026)	D	Saudi Arabia	religious/non-religious tourism demand	quarterly, 2015Q1-2024Q4	none
Hamida and Aloui (2025)	D	Tunisia	tourist arrivals vs geopolitical risk	monthly, 1993-2023	none
Ulucak et al. (2020)	D	Türkiye	international arrivals (gravity)	1998-2017, 25 origins	framing ("sustainable tourism")
Ağazade (2021)	D	Türkiye	tourism revenues	monthly, 2008-01 to 2019-09	none
Alnafisah (2025)	S	GCC	income equality	quarterly, 2014Q1-2023Q4	social
Zmami and Ben-Salha (2024)	S	GCC	three sustainable-development pillars	n/r	economic, social, environmental
Farooq et al. (2023a)	S	GCC	environmental quality	n/r (abstract unavailable)	environmental (title)
Farooq et al. (2023b)	S	GCC (6 countries)	CO2 emissions	annual, 2000-2019	environmental
Harazneh et al. (2026)	S	Jordan	post-COVID recovery index	annual, 2010-2024	environmental (EPI), innovation
Bin Surayhid et al. (2026)	S	Middle East (10 countries)	transport CO2 emissions	annual, 2000-2020	environmental
Voumik et al. (2023)	S	Middle East (10 countries)	CO2 emissions (EKC)	annual, 1997-2019	environmental
Nair and Choudhary (2016)	S	Qatar	sustainable tourism causality	n/r (abstract unavailable)	sustainable tourism (title)
Raihan et al. (2025)	S	Saudi Arabia	carbon neutrality (Hajj)	n/r (abstract unavailable)	environmental (title)
Naseem (2025)	S	Saudi Arabia	CO2 emissions (pilgrimage)	annual, 1996-2022	environmental
Ozturk et al. (2021)	S	Saudi Arabia	CO2 emissions (pilgrimage)	n/r (abstract unavailable)	environmental (title)
?	S	Saudi Arabia	tourism-growth nexus with CO2 emissions (VAR, Granger)	annual, 1990-2018	environmental (CO2 as exogenous variable)
Bildirici and Ersin (2025)	S	Saudi Arabia, Türkiye (+Italy)	pollution, air quality, life expectancy	annual, 1975-2019	environmental, social
Bahar and Demir (2023)	S	Türkiye	CO2 emissions	annual, 1984-2021	environmental
Meimundar and Paris	S	UAE	CO2 emissions (EKC)	annual, 1984-2019	environmental

4.5 Full-Text Evaluation Quality and Risk of Bias

Full texts were appraised for 11 forecasting and 10 econometric studies (Figures 8 and 9).

Forecasting studies. A holdout period was reported in 7 studies (unclear in 3), and 9 used a scale-free or relative accuracy measure. Weaknesses were systematic: only 1 study compared models against a naïve benchmark (Kurtulay and Kızıllırmak, 2024), none tested accuracy differences statistically, 1 reported prediction intervals (Alamoudi and Abed, 2020), and the forecast horizon was stated in 5. Information leakage could be ruled out in only 4 studies; in 3, test-period values of contemporaneous predictors or test-set-based model selection were used (Al Shehhi and Karathanasopoulos, 2018; Alsulami et al., 2026; Bilek, 2025). Table 7 lists the best model and reported accuracy. Machine-learning, neural-network or ensemble models were reported best in 6 of the 11 studies and classical models in 4 (one study fitted a single model). Where comparable, the margins were small, for example a MAPE of 8.06% for an ANN against 8.1% for a Box–Jenkins model (Çuhadar, 2020). Without statistical tests, such rankings do not establish superiority. The single comparison with a naïve benchmark showed a large gain (MAPE 10.63% against 34.41%) (Kurtulay and Kızıllırmak, 2024). Ex-ante forecasts beyond the data were produced in 4 studies (Alamoudi and Abed, 2020; Alsulami et al., 2026; Bilek, 2025; Kayral et al., 2023), none with an environmental or capacity dimension.

Econometric studies. Data sources and periods were always reported and unit-root tests in 8 of 10 studies. Structural breaks were considered in only 2, endogeneity or omitted variables were addressed in 4, and residual diagnostics or stability tests appeared in 4. Extraction also recorded internal reporting inconsistencies, for example coefficients whose signs disagree with their t -statistics (Majumdar and Paris, 2022), a conclusion that contradicts the paper’s own results table (Ouassou and Taya, 2022), conflicting study periods (Wada, 2021) and contradictory statements on the direction of the pilgrim–emission effect (Naseem, 2025).

Direction of emission effects (Theme S). Of the seven full-text studies relating tourism to an environmental indicator (Table 8), two report emission-increasing effects (Majumdar and Paris, 2022; Bildirici and Ersin, 2025), two report decreasing effects (Voumik et al., 2023; Zmami and Ben-Salha, 2024), one reports opposite signs for arrivals (increasing) and receipts (decreasing) (Bahar and Demir, 2023), one is internally ambiguous (Naseem, 2025), and in ? CO₂ enters only as a regressor. No study states whether aviation emissions are included.

	F1 holdout	F2 naïve benchm.	F3 scale-free measure	F4 accuracy test	F5 no leakage	F6 horizon	F7 intervals
AlShehhi (2018)	+	-	+	-	-	-	-
Alamoudi (2020)	-	-	+	-	+	+	+
Alsulami (2026)	+	-	-	-	-	+	-
Bilek (2025)	+	-	+	-	-	+	-
Cuhadar (2020)	?	-	+	-	+	-	-
Kayral (2023)	?	-	+	-	?	+	-
Kurtulay (2024)	+	+	+	-	+	?	-
Laaroussi (2023)	+	-	-	-	?	-	-
Ouassou (2022)	+	-	+	-	?	?	-
Rashad (2022)	+	-	+	-	?	+	-
Tuncsiper (2023)	?	-	+	-	+	-	-

+ Yes (low concern)
- No (high concern)
? Unclear
 Not applicable

Figure 8: Risk of bias and evaluation quality of forecasting studies assessed at full text (checklist items F1–F7, Section 3.7).

	E1 unit root	E2 cointegr.	E3 breaks	E4 CSD / hetero.	E5 endo- geneity	E6 diagn.	E7 data
Alnafisah (2025)	+	?	-	+	?	-	+
Bahar (2023)	+	+	+		-	+	+
Bildirici (2025)	+	+	+		+	-	+
Harazneh (2026)	-	-	-		?	?	+
Jamel (2020)	+	-	-		-	-	+
Majumdar (2022)	+	+	-		+	+	+
Naseem (2025)	+	+	-		?	+	+
Voumik (2023)	+	+	-	+	+	-	+
Wada (2021)	-	?	-	-	-	-	+
Zmami (2024)	+	+	-	+	+	+	+

+ Yes (low concern)
- No (high concern)
? Unclear
 Not applicable

Figure 9: Risk of bias of econometric (demand-modelling and sustainability) studies assessed at full text (checklist items E1–E7).

Table 7: Forecasting studies assessed at full text: best model, reported accuracy and comparison models, as reported by each study (values copied verbatim; scales differ between studies and are not comparable across rows).

Study	Best model (as reported)	Reported accuracy	Comparison models	Ex-ante forecast
Al Shehhi and Karathanasopoulos (2018)	SVM	MAPE 0.0221; MAPE 0.0421	simple moving average, ARIMA	No
Alamoudi and Abed (2020)	ARIMA (2,2,2)	MAPE 9% (train); MAD 180,908 (train)	moving averages MA 2, moving averages MA 3, double moving av	Yes
Alsulami et al. (2026)	VotingR2 ensemble	R2 0.873587 (test)	ARIMAX, Prophet, Decision Tree (baseline)	Yes
Bilek (2025)	SARIMA (1,1,0)(1,1,0)12 (total arrivals)	MAPE 11.06% (test); RMSE 52,508.79 (test)	SARIMA vs SARIMAX (no naive benchmark)	Yes
Çuhadar (2020)	MLP-ANN [4-5-1]	MAPE 8.06%	Holt-Winter's multiplicative-seasonal exponential smoothing,	No
Kayral et al. (2023)	ETS (lowest MAPE and sMAPE for both tourist arrivals and tou	MAPE 10.60% (tourist arrivals)	ETS, ARIMA, TRAMO-SEATS, X13, X11, STL, Grey (single models	Yes
Kurtulay and Kızılrnak (2024)	NAR artificial neural network No. 1 (10 hidden neurons, 15 d	MAPE %10,63 (test)	Naive III, simple moving average, Holt-Winters seasonal expo	No
Laaroussi et al. (2023)	AE-3 layer stacked LSTM	MAE 0.024431 (test); RMSE 0.001169 (test)	RNN, LSTM, PCA-RNN, PCA-LSTM, SVR, ANN, PCA-SVR, PCA-ANN, AE	No
Ouassou and Taya (2022)	LSTM_AR hybrid (ensemble)	MAPE 6.127097 (test)	ARIMA, AR, linear regression, SVR, XGBoost, LSTM	No
Rashad (2022)	Google-augmented model	RMSE 8768.5453 (test); MAE 6279.5498 (test)	traditional model (AR(1) + UK GDP + REER + COVID dummy + mon	No
Tuncsiper (2023)	autoregressive MLP ANN (only model)	MAPE 10.387%; RMSE 0.339	n/r	No

Table 8: Tourism–environment studies assessed at full text: reported effect of tourism on the environmental outcome (coefficients as reported; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ as marked by the authors).

Study	Setting	Estimator	Environmental outcome	Tourism effect on outcome (as reported)	Direction
Bahar and Demir (2023)	Türkiye	ARDL bounds	CO ₂	arrivals +0.247 ($p = 0.0005$); receipts −0.153 ($p = 0.029$), long run	mixed by indicator
Majumdar and Paris (2022)	UAE (Dubai visitors)	ARDL (EKC)	CO ₂	arrivals +0.300*** long run; short run ns	increases
Bildirici and Ersin (2025)	Saudi Arabia, Türkiye, Italy	Fourier VAR (no cointegration)	CO ₂ , PM _{2.5}	lagged arrivals → CO ₂ positive (e.g. Saudi Arabia +0.013**, +0.147**)	increases (short run)
Naseem (2025)	Saudi Arabia (Hajj)	Johansen–VECM	CO ₂	pilgrims: long run reported as negative; short run +0.114 (n/r); no Granger causality	ambiguous (internally inconsistent)
Voumik et al. (2023)	10 Middle East countries	AMG (MG, CCEMG checks)	CO ₂	arrivals −0.012** (AMG); MG ns; CCEMG −0.207*	decreases (weak)
Zmami and Ben-Salha (2024)	GCC (6)	PMG-ARDL	ecological footprint	tourism −0.109** long run; +0.064*** short run	decreases long run
?	Saudi Arabia	VAR in levels	(CO ₂ as regressor only)	no Granger causality CO ₂ ↔ arrivals	not applicable

Table 9: Counts of included studies by forecasting objective and sustainability content (abstract-level coding).

	Sustainability dimension present	Absent
Forecasts tourism outcome (F)	1	15
Does not forecast (D, S)	18	8

Table 10: Sensitivity analyses of the forecasting–sustainability split.

Analysis	Studies	Forecasting	... with sustainability	Non-forecasting	... projecting forward
Main analysis	42	16	1	26	1
S1: plus growth-nexus exclusions	51	16	1	35	1
S2: without Türkiye	32	10	1	22	1
S3: full-text-confirmed only	21	11	0	10	0

4.6 Cross-Theme Synthesis: The Forecasting–Sustainability Split

Cross-classifying the 42 studies by whether they forecast and whether they include a sustainability dimension (stated environmental, social, economic-sustainability or climate content, excluding mere framing) gives Table 9. Only one study sits in the forecasting-with-sustainability cell (Louati et al., 2024), and its sustainability link is stated at the level of implications rather than constrained forecasting. Two further forecasting studies mention sustainability only as framing: Bilek (2025) in its title and Zamzami (2026) in its policy implications. The 18 sustainability-bearing non-forecasting studies explain past relationships; with the exception noted above they do not project the pressures that future arrivals will create.

Sensitivity analyses. The split is robust (Table 10). Adding back the growth-nexus exclusions (S1), removing Türkiye (S2) or restricting to full-text-confirmed studies (S3) leaves at most one forecasting study with a sustainability dimension and at most one non-forecasting study that projects forward.

Two keyword checks were made on the 34 available abstracts. No abstract mentions water or waste, and none quantifies a capacity limit; only Alamoudi and Abed (2020) mentions utility and infrastructure capacity, as a policy recommendation. Prediction intervals or probabilistic forecasts are not named in any forecasting abstract, although absence from an abstract is weaker evidence than absence from the full text.

4.7 Reporting Quality of Forecasting Studies (Abstract Level)

Among the 16 forecasting studies, 14 had abstracts permitting assessment. 13 of these reported a benchmark comparison, 13 named an accuracy metric, and 6 produced forward projections (the same studies need not overlap). Four forecasting studies had no abstract in the bibliographic record and were not assessable. We caution that these proportions describe abstract-level reporting, not methodological quality.

5 Discussion

5.1 Principal Findings

The retrieved Middle Eastern evidence base is recent, concentrated in Saudi Arabia and Türkiye, and split into two traditions that rarely meet. Forecasting studies employ increasingly modern

machine-learning pipelines but target arrivals, spending or prices; sustainability studies use ARDL-type designs to explain emissions or development outcomes but, with one scenario-based exception, do not project forward. The links also run in different directions: sustainability studies ask how tourism affects the environment, the one scenario study asks how climate affects tourism, and no included study links forecast visitor numbers to future environmental load. This split, rather than any single method comparison, is the main insight of the review. It is consistent with the broader forecasting literature’s focus on accuracy against benchmarks (Makridakis et al., 2020, 2022), which provides no mechanism for taking sustainability limits into account.

5.2 Comparison with Prior Reviews

Several regional findings agree with the global reviews, and some depart from them. The global reviews report the rise of AI and hybrid models (Jiao and Chen, 2018); in this corpus, 6 machine- or deep-learning and 5 hybrid or comparative designs make up most of the 16 forecasting studies, which is consistent. Search-engine and internet data were identified as an emerging trend globally (Wu et al., 2024), yet only 3 of the 16 regional forecasting abstracts use such data (Rashad, 2022; Laaroussi et al., 2023; Bilek, 2025), which suggests a lag in adoption. The conflicting tourism–emission results reported globally by Sun et al. (2022) recur regionally. Among the seven full-text Theme S studies with an environmental indicator, effects split two increasing, two decreasing, one mixed by tourism indicator and one ambiguous (Section 4.5), and arrivals and receipts carry opposite signs within the same model in both Türkiye (Bahar and Demir, 2023) and GCC data (Farooq et al., 2023b). Sun et al. (2022) also note that national-level studies exclude international aviation emissions; none of the regional studies reports whether aviation is included. This matters especially for long-haul and pilgrimage travel to Saudi Arabia. The forecasting results parallel the global forecasting literature’s emphasis on benchmark-based, out-of-sample evaluation (Makridakis et al., 2020; Hyndman and Athanasopoulos, 2021). That standard is rarely met here: one study used a naïve benchmark and none used a formal accuracy test. Relative to Saleh et al. (2021), who found GCC tourism research to focus mostly on Dubai, Saudi Arabia is the most frequent setting here (13 of 42 studies, all from 2020 onward), while only 2 included studies concern the UAE specifically. Differences in scope and inclusion criteria mean these contrasts describe the two corpora rather than establishing a trend.

5.3 Theoretical Implications: A Framework of Tourism–Environment Links

Figure 10 organises the included studies by the direction of the link they examine. Theme F studies model drivers $\rightarrow$ demand and project demand forward; Theme S studies estimate tourism $\rightarrow$ environment relationships ex post; one Theme D study (Chouari, 2025) projects climate $\rightarrow$ demand under scenarios. The link from *forecast* demand to future environmental load or capacity is absent from the corpus.

Capacity-aware forecasting. We propose, as a *hypothesised direction that has not been tested here*, that tourism forecasts for sustainability-sensitive destinations be evaluated not only by point accuracy but by the decision-relevant risk of exceeding capacity. Let $Y_{r,t+h}$ be visitors to region r at horizon h , with predictive distribution $\hat{F}_{r,t+h}$ and environmental capacity K_r (for example water, energy or site-carrying thresholds). The relevant quantity is the exceedance probability $\hat{p}_{r,t+h} = 1 - \hat{F}_{r,t+h}(K_r)$, assessed with proper scoring rules and calibration diagnostics. Such an agenda would require probabilistic forecasts, regionally disaggregated data and environmental indicators, none of which is combined in the included studies. The Saudi giga-project and heritage destinations make this question concrete, but testing it requires data not examined in this review. This agenda is not new in its aim: carrying-capacity research already builds early-warning indicator systems and forecasts them, for example with neural networks (Ye et al., 2020), and Long et al.

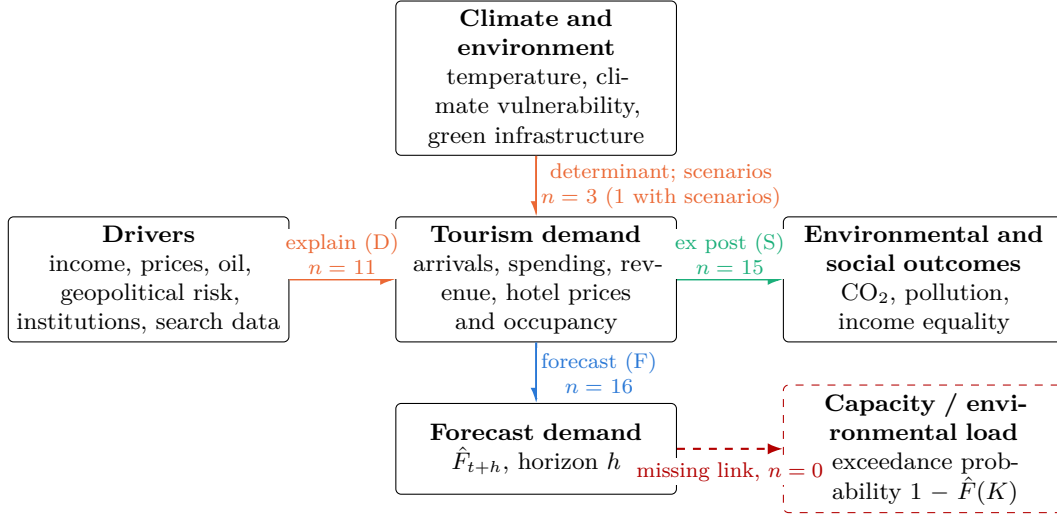


Figure 10: Framework of the tourism–environment links examined by the included studies. Solid arrows are links studied (number of studies); the dashed arrow is the missing link proposed for future research.

(2022) call for dynamic forecasting of carrying capacity. The difference we propose is to derive warnings from probabilistic *demand* forecasts, evaluated for calibration, rather than to forecast a composite warning index. Whether this improves decisions is an empirical question.

5.4 Practical Implications

For researchers: report accuracy metrics, benchmarks and forecast horizons in abstracts, and compare forecasts using accepted predictive-accuracy tests (Diebold and Mariano, 1995). For policy: forecasts used for Vision 2030 planning should be accompanied by environmental-capacity analysis; the existing nexus studies report tourism–emissions links in GCC and UAE data, with signs that differ across tourism indicators and sustainability pillars (Farooq et al., 2023b; Zmami and Ben-Salha, 2024; Majumdar and Paris, 2022).

5.5 Future Research Agenda

Table 11 lists research directions that follow from the gaps identified, each linked to the evidence in this review.

6 Conclusions

6.1 Limitations

1. **Partial full-text assessment.** The execution environment blocked publisher and DOI-resolver hosts. Full-text eligibility was checked for 22 of 43 provisionally included studies (one excluded); 21 remain included on title-and-abstract evidence, and data extraction for all studies rests on abstracts and Crossref metadata. 8 included records had no abstract. Extracted items marked n/r were never inferred.
2. **Limited databases, capped retrieval.** Crossref (top 100 records per query) supplemented by Semantic Scholar; OpenAlex was unavailable because of quota limits, and Scopus, Web of Science, IEEE Xplore, Arabic-language and regional databases and grey literature were not searched (planned amendment).

Table 11: Future lines of research.

Research line	Rationale from this review
Probabilistic forecasts and prediction intervals	No forecasting abstract names prediction intervals or probabilistic forecasts (Section 4.6).
Capacity-aware evaluation of demand forecasts	No study links forecast demand to environmental load or capacity (Figure 10); early-warning research forecasts indices instead (Ye et al., 2020; Long et al., 2022).
Forecast-based emission projections	Sustainability studies estimate tourism–emission elasticities ex post (15 studies) but none projects emissions from forecast arrivals.
Saudi regional and event-based forecasting	Only 4 Saudi forecasting studies; regional spending varies strongly across 13 regions (Alfehaid, 2026); religious tourism has distinct dynamics (Mabrouk and Alanzi, 2026).
Evaluation standards	Accuracy measures appear in only 13 of 14 assessable abstracts (Figure 7); benchmark and test reporting (Hyndman and Athanasopoulos, 2021; Diebold and Mariano, 1995).
Search and big data	Only 3 of 16 forecasting studies use search data, against a global trend (Wu et al., 2024).

3. **Imperfect automated screening.** A recall check against 6 supplementary records found that two had been retrieved but removed by the automated filters and four had not been retrieved at all. The pipeline therefore missed relevant studies; a point estimate of pipeline recall over included studies is $37/42 \approx 88\%$, which is an upper bound because the check was seeded from the same search family.
4. **Single reviewer.** Screening was performed by one AI reviewer, and full-text extraction by AI agents with spot checks. A blinded pack for an independent human reviewer has been prepared, but agreement (κ) is not yet available. One screening error was found and corrected by a second-pass check (Section 3.5).
5. **Partial risk-of-bias assessment.** Risk of bias was appraised for the 21 full-text studies with adapted checklists that have not been validated; reporting-bias and certainty assessments (PRISMA items 14, 15, 21, 22) and effect-size pooling were not performed.
6. **Scope choices.** The definition of Middle East includes Türkiye and Iran; the exclusion of growth-only nexus studies is a judgement; the corpus includes economic-growth-adjacent sustainability outcomes only where an environmental, social or sustainability dimension was stated.
7. **Keyword-based sustainability coding** reflects abstract wording and may overlook relevant content.

Because of limitations 1–4, the review should be read as a systematic mapping of a retrievable subset of the literature, and the paper requires full-text verification and, ideally, a second reviewer before submission to a journal.

6.2 Contributions

The review contributes a joint map of forecasting and sustainability evidence for Saudi Arabia and the Middle East, a quantified forecasting–sustainability split supported by both thematic coding and keyword co-occurrence, a framework of link directions, and a research agenda positioned against prior reviews and early-warning research.

6.3 Concluding Remarks

Within the retrievable Middle Eastern literature (42 studies), tourism forecasting and tourism-sustainability analysis are largely separate: forecasting studies seldom include sustainability and sustainability studies rarely project forward. Saudi Arabia has a growing but thin forecasting base. A capacity-aware, probabilistic forecasting agenda could connect the two. Verification against full texts, a second reviewer and additional databases are the necessary next steps.

Declarations

Registration and protocol: not registered (see Methods). **Funding:** none declared [to be completed]. **Competing interests:** none declared [to be completed]. **AI assistance disclosure:** the search code, screening, data coding and drafting were performed by an AI agent (Claude Code), supervised by the author; the human author must verify all claims and citations before submission. **Data and code availability:** all search logs, screening outputs, coded data and scripts are in the repository directory `review/` (branch `claude/quirky-goldberg-12lm00`); records can be regenerated by running `search/crossref_search.py`, `screen.py`, `screen_stage2.py`, `build_dataset.py` and `gen_tex.py`.

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A PRISMA 2020 checklist with item status

The completed official PRISMA 2020 checklist (main and abstract checklists) is supplied as a supplementary file (`prisma/PRISMA_2020_Checklist_completed.docx`); the summary below gives the status of each item.

Item	Topic	Status	Where / comment
1	Title	Reported	Title identifies a systematic review.
2	Abstract	Reported	Structured; PRISMA 2020 for Abstracts elements.
3	Rationale	Reported	Sections 1–2.
4	Objectives	Reported	RQ1–RQ4, Section 1.
5	Eligibility criteria	Reported	Section 3.4, Table 3.
6	Information sources	Partial	Crossref only; date stated (Section 3.2).
7	Search strategy	Reported	Section 3.3, Table 2, Appendix C; PRISMA-S not fully met.
8	Selection process	Partial	Single AI reviewer; full-text check of 21 of 42 reports; no duplicate (Section 3.5).
9	Data collection process	Partial	Single reviewer, abstract-level (Section 3.6).
10a/b	Data items	Reported	Section 3.6, Table 4.
11	Study risk of bias assessment	Partial/Not done	Reporting-transparency screen only (Section 3.7).
12	Effect measures	Not applicable	No pooling.
13a–f	Synthesis methods	Reported	Descriptive, keyword and thematic analysis (Section 3.8).
14	Reporting bias assessment	Not done	–
15	Certainty assessment	Not done	–
16a/b	Study selection	Reported	Figure 1, Section 4.1, Appendix E; completed PRISMA template supplied.
17	Study characteristics	Reported	Figures 2–4, Tables 5–6.
18	Risk of bias in studies	Partial	Section 4.6.
19	Results of individual studies	Partial	Abstract-level only.
20a–d	Results of syntheses	Partial	Sections 4.2–4.5; no meta-analysis.
21	Reporting biases	Not done	–
22	Certainty of evidence	Not done	–
23a–d	Discussion	Reported	Sections 5–6.
24a–c	Registration and protocol	Not done	Not registered.
25	Support	Reported	None declared [to be completed].

Item	Topic	Status	Where / comment
26	Competing interests	Reported	None declared [to be completed] .
27	Availability of data/code	Reported	Repository review /.

B Coding scheme

Theme: F, D or S (Section 3.4). Method family: C (classical statistical or econometric), M (machine or deep learning), H (hybrid, ensemble or multi-family comparison, e.g. classical versus neural models), G (grey). Benchmark: abstract names a comparison of models or benchmarks (y), none (n), unknown (n/r). Forward projection: a forecast beyond the sample or scenarios stated in the abstract. Sustainability: stated dimension; “framing” denotes the word “sustainable” used without a sustainability variable.

C Search log

Query (<code>query.bibliographic</code>)	Reported total	Retrieved
tourism demand forecasting Saudi Arabia	412,893	100
tourist arrivals forecasting Saudi Arabia	206,248	100
tourism forecasting machine learning Saudi Arabia	1,685,238	100
tourism forecasting deep learning Saudi Arabia	1,771,597	100
hotel occupancy forecasting Saudi Arabia	218,316	100
tourism sustainability prediction Saudi Arabia	894,148	100
tourism CO2 emissions Saudi Arabia	423,634	100
sustainable tourism modelling Saudi Arabia	884,515	100
tourism demand forecasting Middle East	630,664	100
tourist arrivals forecasting Middle East	424,968	100
tourism forecasting machine learning Middle East	1,899,471	100
tourism forecasting deep learning Middle East	1,985,114	100
hotel occupancy forecasting Middle East	437,099	100
tourism sustainability prediction Middle East	1,111,210	100
tourism CO2 emissions Middle East	641,367	100
sustainable tourism modelling Middle East	1,101,189	100
tourism demand forecasting GCC	319,454	100
tourist arrivals forecasting GCC	112,281	100
tourism forecasting machine learning GCC	1,594,056	100
tourism forecasting deep learning GCC	1,680,493	100
hotel occupancy forecasting GCC	124,380	100
tourism sustainability prediction GCC	801,859	100
tourism CO2 emissions GCC	330,019	100
sustainable tourism modelling GCC	792,073	100
tourism demand forecasting Hajj Umrah	323,426	100
tourist arrivals forecasting Hajj Umrah	116,271	100
tourism forecasting machine learning Hajj Umrah	1,597,894	100
tourism forecasting deep learning Hajj Umrah	1,684,325	100
hotel occupancy forecasting Hajj Umrah	128,367	100
tourism sustainability prediction Hajj Umrah	805,908	100
tourism CO2 emissions Hajj Umrah	334,023	100
sustainable tourism modelling Hajj Umrah	796,103	100
tourism demand forecasting Gulf Cooperation Council	426,549	100
tourism sustainability Gulf Cooperation Council	553,013	100
tourism demand forecasting GCC	319,454	100
tourism sustainability GCC	446,420	100

Query (<code>query.bibliographic</code>)	Reported total	Retrieved
tourism demand forecasting Middle East	630,664	100
tourism sustainability Middle East	756,893	100
tourism demand forecasting MENA	331,865	100
tourism sustainability MENA	458,825	100
tourism demand forecasting Dubai	320,131	100
tourism sustainability Dubai	447,184	100
tourism demand forecasting UAE	323,571	100
tourism sustainability UAE	450,488	100
tourism demand forecasting United Arab Emirates	549,400	100
tourism sustainability United Arab Emirates	675,840	100
tourism demand forecasting Qatar	323,738	100
tourism sustainability Qatar	450,720	100
tourism demand forecasting Oman	328,493	100
tourism sustainability Oman	455,487	100
tourism demand forecasting Bahrain	319,661	100
tourism sustainability Bahrain	446,731	100
tourism demand forecasting Kuwait	323,597	100
tourism sustainability Kuwait	450,655	100
tourism demand forecasting Jordan	362,023	100
tourism sustainability Jordan	488,890	100
tourism demand forecasting Egypt	388,672	100
tourism sustainability Egypt	515,428	100
tourism demand forecasting Turkey	382,632	100
tourism sustainability Turkey	509,550	100
tourism demand forecasting Iran	426,257	100
tourism sustainability Iran	553,094	100
tourism demand forecasting Morocco	336,617	100
tourism sustainability Morocco	463,578	100
tourism demand forecasting Tunisia	326,643	100
tourism sustainability Tunisia	453,654	100
tourism demand forecasting Lebanon	323,843	100
tourism sustainability Lebanon	450,895	100
tourism demand forecasting Israel	351,835	100
tourism sustainability Israel	478,874	100

D Records and code

Raw Crossref records (`data/crossref_raw.json`), the log (`data/search_log.json`), screening outputs, the coded dataset (`data/included.json`), and the PRISMA counts (`data/final_numbers.json`) are in the repository. Web-search leads from the earlier screening passes are in `data/stream*.json`; they are unverified and were used only to seed the targeted searches.

E Records excluded at eligibility assessment

DOI	Title (truncated)	Year	Reason
10.52133/ijrsp.v6.71.6	Enhancing Risk Management in Events by Providing AI-Risk Assessment Framework	2025	Not quantitative forecast/demand/sustainability study
10.18089/tms.2022.180302	Effect of internal corporate social responsibility activities on tourism and hospitality employ	2022	Not quantitative forecast/demand/sustainability study
10.30892/gtg.634spl28-1643	THE TRANSFORMATIVE ROLE OF PLACE ATTACHMENT IN CONVERTING SUSTAINABILITY PERCEPTIONS INTO ACTIV	2025	Not quantitative forecast/demand/sustainability study

DOI	Title (truncated)	Year	Reason
10.3727/108354215x1446484587095	Religious Tourism and Economic Growth in Oil-Rich Countries: Evidence from Saudi Arabia	2015	Growth nexus only
10.18488/31.v12i2.4595	Hybrid tourism and territorial development: The case of voluntourism and alternative tourism in	2025	Not quantitative forecast/demand/sustainability study
10.3390/economies9030117	The Role of Tourism in Economic Growth: Empirical Evidence from Saudi Arabia	2021	Growth nexus only
10.21833/ijaas.2021.09.008	COVID-19 and tourism sector: A perspective of Saudi Arabia	2021	Not quantitative forecast/demand/sustainability study
10.30892/gtg.662spl37-1798	THE INFLUENCE OF THE DIVERSITY OF TOURIST REGIONS ON THE CHOICE OF TOURIST DESTINATIONS IN SAUD	2026	Not quantitative forecast/demand/sustainability study
10.3390/su132313372	Forecasting the Impact of Gross Domestic Product (GDP) on International Tourist Arrivals to Lan	2021	Outside Middle East
10.30519/ahtr.990903	Forecasting Tourist Arrivals to Sangiran Using Fuzzy with Calendar Variations	2022	Outside Middle East
10.1108/jhti-03-2024-0223	Enabling hotel circularity via Industry 4.0 innovations for enhanced hotel performance: insight	2024	Not quantitative forecast/demand/sustainability study
10.3390/su17052328	Leveraging Spectral Clustering and Long Short-Term Memory Techniques for Green Hotel Recommenda	2025	Not quantitative forecast/demand/sustainability study
10.30892/gtg.43321-915	THE IMPACT OF RELIGIOUS TOURISTS' SATISFACTION WITH HAJJ SERVICES ON THEIR EXPERIENCE AT THE SA	2022	Not quantitative forecast/demand/sustainability study
10.3390/su18063124	Sustainable Destination Marketing in Saudi Arabia: Effects of Sustainability Storytelling and S	2026	Not quantitative forecast/demand/sustainability study
10.55908/sdgs.v11i11.1639	Challenges Facing Tourism for People with Disabilities (Accessible Tourism) in the Kingdom of S	2023	Not quantitative forecast/demand/sustainability study
10.24922/eot.v6i2.53474	Motivation towards Inbound Tourism: A Study of Middle East Tourist	2019	Not quantitative forecast/demand/sustainability study
10.63556/jotags.2026.1930	Türkiye'ye Yönelik Yabancı Turizm Talebinin Öngörülmesi: Reel Efektif Döviz Kuru, Mevsimsellik	2026	Non-English / proceedings
10.14505/jemt.v14.4(68).03	The Contribution of Tourism Industry Cash Flows to Economic Growth "Middle East Evidence"	2023	Growth nexus only
10.24114/cess.v7i2.33959	Forecasting Room Occupancy Rates Based on Hotel Class in Bali Using the ARIMA Method	2022	Outside Middle East
10.3390/su12229620	Urban Sprawl, Socioeconomic Features, and Travel Patterns in Middle East Countries: A Case Stud	2020	Not quantitative forecast/demand/sustainability study
10.32479/ijeeep.14410	Do Tourism and Renewable Energy Influence CO2 Emissions in Tourism-Dependent Countries?	2023	Region/tourism link unconfirmed
10.1371/journal.pone.0328456	Estimating CO2 emissions due to present and future suborbital space tourism industry	2025	Outside Middle East
10.3390/su8070605	Is Tourism Development a Sustainable Economic Growth Strategy in the Long Run? Evidence from GC	2016	Growth nexus only
10.3390/su162310760	Transport Sector Emissions and Environmental Sustainability: Empirical Evidence from GCC Econom	2024	Region/tourism link unconfirmed
10.3390/su17115058	Sustainable Tourism and Its Environmental and Economic Impacts: Fresh Evidence from Major Touri	2025	Outside Middle East
10.29313/amwaluna.v8i1.3476	Bibliometric Analysis of Halal Tourism (Hajj and Umrah) Development in the World	2024	Not quantitative forecast/demand/sustainability study
10.47747/ijmhrr.v5i2.1705	The Influence of Service Quality, Prices, and Promotions on the Satisfaction of Pilgrims on Haj	2024	Not quantitative forecast/demand/sustainability study
10.55677/gjefr/02-2025-vol02e10	Shariah-Compliant Financial Services for Hajj and Umrah: Products, Challenges, and Policy Recom	2025	Not quantitative forecast/demand/sustainability study
10.1109/access.2025.3532385	Machine Learning-Integrated Usability Evaluation and Monitoring of Human Activities for Individ	2025	Operational/crowd/health prediction
10.1016/j.aej.2025.12.032	Predictive analysis of Hajj and Umrah performance using key performance indicators (KPIs) and m	2026	Operational/crowd/health prediction
10.61132/wjilt.v2i3.434	Digital Books as Learning Media for Islamic Religious Education on Hajj and Umrah Material for	2025	Not quantitative forecast/demand/sustainability study
10.1016/j.eswa.2023.122204	A customized deep learning-based framework for classification and analysis of social media post	2024	Not quantitative forecast/demand/sustainability study

DOI	Title (truncated)	Year	Reason
10.21580/nw.2025.19.1.26704	Building Virtual Reality Hajj Journey Learning Media as a Simulation Tool for Teaching Hajj an	2025	Not quantitative forecast/demand/sustainability study
10.38035/jgia.v3i4.568	Economic Improvement Based on Hajj and Umrah Business in Indonesia	2025	Not quantitative forecast/demand/sustainability study
10.1109/access.2024.3394905	Crowd Congestion Forecasting Framework Using Ensemble Learning Model and Decision Making Algori	2024	Operational/crowd/health prediction
10.5339/jist.2025.12	Utilizing information technology to address challenges in Hajj and Umrah: A narrative review	2025	Not quantitative forecast/demand/sustainability study
10.55041/ijrem37043	3D Modelling for the Hajj and Umrah Pilgrims	2024	Not quantitative forecast/demand/sustainability study
10.1016/j.aej.2025.06.043	Smart mobile application for real-time monitoring and prediction of pilgrim crowd activities in	2025	Operational/crowd/health prediction
10.24191/idealogy.v10i2.834	An Analysis on 3D Modelling Technique in Virtual Reality for Hajj and Umrah Simulation	2025	Not quantitative forecast/demand/sustainability study
10.1080/1528008x.2025.2599751	PLS – Neural Network Approach to Study Factors Influencing Gulf Cooperation Council Tourists’	2025	Not quantitative forecast/demand/sustainability study
10.53596/ma0frs79	Regional Tourism and Global Challenges in the Face of Political Uncertainty: Case STUDY of the	2025	Not quantitative forecast/demand/sustainability study
10.14505//jemt.v12.2(50).29	The Contribution of Tourism to Economic Growth: The Case of Qatar	2021	Growth nexus only
10.47259/ijrebs.2025.634	Revitalization of Oman’s Tourism Destinations through the Omani Youth’s Level of Engagement and	2025	Not quantitative forecast/demand/sustainability study
10.3727/154427320x1582879251	An Analytical Approach to Cruise Tourism As an Option For Development: a Case Study of the Sult	2020	Not quantitative forecast/demand/sustainability study
10.3390/admsci12040168	Fusion of Sustainability in the Tourism Industry for Improved Competitiveness: Investigation of	2022	Not quantitative forecast/demand/sustainability study
10.3390/su11071907	Residents’ Perceptions and Satisfaction toward Tourism Development: A Case Study of Petra Regio	2019	Not quantitative forecast/demand/sustainability study
10.30892/gtg.50403-1121	PERCEPTION OF LOCAL COMMUNITY TOWARDS TOURISM DEVELOPMENT: A STUDY ON RURAL TOURISM SITES OF JO	2023	Not quantitative forecast/demand/sustainability study
10.30892/gtg.58105-1390	TOURISM EXPERIENCES, MOTIVATIONS, AND TRAVEL LIFESTYLES PREFERENCES FOR DOMESTIC TOURISTS: A CA	2025	Not quantitative forecast/demand/sustainability study
10.3390/su151310013	Determinants of Residents’ Support for Sustainable Tourism Development: An Empirical Study in M	2023	Not quantitative forecast/demand/sustainability study
10.18092/ulikidince.442734	RELIGION AND TOURISM IN TURKEY: AN ECONOMICALLY EMPIRICAL STUDY RELIGION AND TOURISM IN TURKEY:	2019	Growth nexus only
10.31201/ijhmt.696185	AGRO TOURISM IN TURKEY	2020	Not quantitative forecast/demand/sustainability study
10.34190/eckm.27.1.4788	Hybrid ARIMA-LSTM Knowledge Management for Strategic Tourism Demand Forecasting in Morocco	2026	Non-English / proceedings
10.55677/gjefr/10-2024-vol01e7	VECM Model Analysis of the Causality between Tourism, Foreign Direct Investment and Economic Gr	2024	Growth nexus only
10.1177/1354816619836338	Tourism–growth nexus under duress: Lebanon during the Syrian crisis	2019	Growth nexus only
10.30892/gtg.27417-436	THE ROLE OF RELIGIOUS TOURISM IN SUSTAINABLE DEVELOPMENT IN SAUDI ARABIA: EVIDENCE FROM QUANTIL	2019	Full text: growth nexus only
10.1016/j.ijhm.2026.104745	Hybrid machine learning approaches for hotel occupancy forecasting: Evaluating gradient boostin	2026	Region/tourism link unconfirmed